VEGA LEDGER

Autonomous Options Desk & Cryptographic Proof Engine

Engineered around formal Deflated Sharpe Ratio (DSR) mathematical falsification, deterministic capital preservation gates, and live Ethereum Sepolia L1 audit trails.

OFFICIAL ACCOUNT ID	STARTING & CURRENT CAPITAL	L1 ANCHOR STATUS
PA3LL11TFH7L	$100,000.00 Cash (0.00% DD)	Ethereum Sepolia Verified

Most AI Trading Bots Hallucinate Alpha

Over 150 hackathon submissions claim to have "predictive trading agents." In reality, they fall into three fatal quantitative pitfalls that wipe out institutional capital.

01. Unconstrained LLM Consensus

Multi-agent debates (Bull vs. Bear) suffer from emotional groupthink (ICLR 2026). LLMs generate confident directional bets that amount to coin flips, lacking any mathematical expectation of positive return.

02. Frictionless Backtest Fallacy

Standard backtests assume mid-price fills with zero bid-ask slippage. In live options markets, spread friction ($0.48 round trip per contract) consumes >28% of credit, turning winning models negative.

03. Forced Trading in Low-IV

Retail bots feel pressured to "stay busy," blindly selling premium when SPY IV Rank is <10%. Selling options at rock-bottom prices carries asymmetric tail risk for pennies of return.

Mathematical Falsification: We Rejected Our Own Thesis

Instead of p-hacking parameters to present a fraudulent win, we subjected our core SPY Volatility Risk Premium (VRP) condor strategy to institutional Deflated Sharpe Ratio (DSR) testing.

19-Year Backtest vs. Real-World Friction (2007–2026)

STRATEGY SPECIFICATION	GROSS SHARPE	NET SHARPE ($0.48/SH)	DEFLATED SHARPE (DSR)
Unconditional $10-Wide Condor	+1.62	-0.05	0.00% (REJECTED)
Regime-Filtered $20-Wide Condor	+0.85	-0.24	0.00% (REJECTED)

Key Rigor Takeaways

- **10,000 Stationary Bootstrap Splits:** 95% Confidence Interval straddles zero: [-1.33, +1.00].
- **Zero Statistical Significance:** DSR=0.00% proves the gross Sharpe is an artifact of selection bias and market beta.
- **The Institutional Mandate:** Because the strategy has negative expectancy net of costs, the agent **refuses to trade it.** Zero alpha claimed without proof.

Autonomous Architecture: Code Overrides AI

Vega Ledger separates analytical reasoning from execution authority. Built on LangGraph, Alpaca's Trading API, and FastMCP, deterministic Python holds unconditional veto power.

LangGraph State Machine

Modular directed acyclic graph (DAG). Ingests real-time market telemetry, evaluates IV surfaces, schedules single-leg and multi-leg option orders, and handles broker reconciliation.

Alpaca Trading API & MCP

Direct integration with Alpaca's modern Python SDK and FastMCP server. Evaluates option chains, bid-ask quotes, Greeks, and executes structured orders with fail-closed exception handling.

Terminal Workstation UI

Dense Bloomberg-style Next.js 16 terminal. Live polling of account equity, buying power, risk telemetry, open orders, and verifiable decision block inspectors on port 3001.

Deterministic Risk Gates: Protecting Principal First

Every proposed order must clear three mathematical filters before reaching Alpaca's broker endpoint (`src/risk/gate.py`). If any test fails, the agent stands down.

1. Volatility Regime Gate

Monitors SPY IV Rank, VIX 200-day moving average, and 5-day velocity. Mandates **IV Rank > 25%** to sell options. When IV Rank dropped to 4.8% this week, the gate locked closed.

2. Capital Sizing Limiter

Strict non-bypassable constraints: Max 2% of portfolio equity risked per trade. Total portfolio allocation capped at 25% below VIX 22, 15% at VIX 22–30, and 5% above VIX 30.

3. The ML Disarmament

An experimental Random Forest momentum model predicted SPY direction. Because it lacked walk-forward DSR proof, the risk governor **benched the model** and refused live execution.

Cryptographic Proof: Anchored to Ethereum L1

Most trading bots keep private, editable text logs. Vega Ledger commits every decision to an immutable SHA-256 hash chain and anchors the state to the public Ethereum blockchain.

On-Chain Verification Telemetry

NETWORK: Ethereum Sepolia (Chain ID: 11155111)

TRANSACTION: 0x46f3ce0bdb9e343a4d2c85b57146a5c2fca5c5dd861c9aadab00f21bb75f396d

ROOT HASH: c54f1a609a82a5d6009ad12931b41d73cd4ba4ed7fddd5aa1c4cfc5d14660c1a

STATUS: Confirmed on Etherscan (100% Immutable)

Zero Post-Hoc Revisionism

Each decision block hashes the previous block's hash, market telemetry, and risk verdicts.

If an operator tampers with a single byte of past trade history or alters a refusal retroactively, the cryptographic chain loudly breaks and fails local verification (`python verify_log.py`).

Tail-Risk Defense: Outperforming in Crises

Per Alpaca's judging FAQ guidelines on flat market conditions, our regime filters were validated across the three largest volatility shocks of the 21st century.

2008 Financial Crisis

-1.09%

While the S&P 500 collapsed **-47.17%** and unhedged options sellers suffered ruin, our regime filter cut allocation and preserved 98.9% of capital.

2018 Volmageddon

-2.13%

XIV inverse-VIX products liquidated in 24 hours (SPY -19.20%). The regime filter detected VIX velocity and halted new writes within 15 minutes.

2020 COVID Crash

-2.14%

Fastest 30% drop in market history (SPY -33.72%). Vega Ledger's circuit breakers locked trading, preventing catastrophic wing breaches.

Scoring Window: 100% Capital Preserved

In options trading, market conditions dictate opportunity. When expected value is negative, sitting in cash is the only winning quantitative move.

Official Competition Posture

ACCOUNT ID: **PA3LL11TFH7L**
STARTING CAPITAL: $100,000.00 Cash
FINAL EQUITY: **$100,000.00 (0.00% Drawdown)**
POSTURE: 100% Cash Reserves (Zero Open Risk)
SPY IV RANK: 4.8% (Entry Hurdle: >25.0%)

Why Vega Ledger Wins

- **Mathematical Falsification:** Rejected our own strategy via DSR (0.00%). Zero alpha claimed without proof.
- **Live L1 Blockchain Proof:** Broadcasted tamper-proof root hash to Ethereum Sepolia.
- **Institutional Discipline:** Refused to gamble principal into an uncompensated 4.8% IV market.
- **Complete Codebase:** LangGraph state machine, FastMCP, live Next.js terminal workstation.